

Audio Description for Educational Videos on COVID-19 Response: A Corpus-Based Study on Linguistic and Textual Idiosyncrasies

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Audio description (AD), unlike interlingual translation and interpretation, is subject to unique constraints as a spoken text. Facilitated by AD, educational videos on COVID-19 anti-virus measures are made accessible to the visually disadvantaged. In this study, a corpus of AD of COVID-19 educational videos is developed, named “Audio Description Corpus of COVID-19 Educational Videos” (ADCCEV). Drawing on the model of Textual and Linguistic Audio Description Matrix (TLADM), this paper aims to identify the linguistic and textual idiosyncrasies of AD themed on COVID-19 response released by the New Zealand Government. This study finds that linguistically, the AD script uses a mix of complete sentences and phrases, the majority being in Present Simple tense. Present participles and the “with” structure are used for brevity. Vocabulary is diverse, with simpler words for animated explainers. Third-person pronouns are common in educational videos. Color words are a salient feature of AD, where “yellow” denotes urgency, and “red” indicates importance, negativity, and hostility. On textual idiosyncrasies, coherence is achieved through intermodal components that align with the video’s mood and style. AD style varies depending on the video’s purpose, from informative to narrative or expressive.

Keywords: Audio Description, COVID-19, educational videos, corpus-based study

Introduction

As an emerging modality of audiovisual translation (AVT), Audio Description (AD) is an assistive verbal commentary of visual representations in a variety of multimodal forms for accessible, inclusive purposes. AD can be applied in a wealth of scenarios, including theaters, museums, churches, and other multimodal media (i.e. posters, movies, and short videos).

The end-users or beneficiaries of AD are diverse. Originally designed to make audiovisual products available for blind and visually impaired persons (VIPs) (Perego, 2018), AD also provides benefits for individuals with good vision. For those who need assistance keeping track of tale elements or are unable to keep their eyes on the screen because they are engaged in another activity, such as cooking, AD offers access to video content (Audio Description, n.d.).

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Different from interlingual translation and interpretation products, AD is subject to certain idiosyncratic constraints, linguistically and textually, due to its nature as a text to be spoken. To relay information embedded sophisticatedly in a non-linear manner, AD is developed with words arrayed in a linear way after careful selection of content and lexical and syntactical consideration of internal and external cohesion and coherence (Mazur, 2022).

The COVID-19 pandemic is the most widespread outbreak that has afflicted the world in the past century. The pandemic, a severe crisis and a formidable challenge, posed a significant danger to human life and health. Educational videos on anti-virus measures in the context of COVID-19 are supposed to be accessible to both the general public and the most vulnerable, the VIPs in particular. Facilitated by AD, both people with vision loss and those with good vision have access to the educational content on COVID-19 response, thus health tips can reach every individual who receives the multimodal messages from these audiovisual products.

Despite the ready-made guidelines for AD in European and North American countries (Rai et al., 2010) and the current understanding of AD characteristics for general materials (such as films), investigations into AD in educational videos on COVID-19 are still rare. To explore the linguistic and textual idiosyncrasies of AD, corpus-based investigations provide feasible and effective solutions.

This paper aims to identify the linguistic and textual idiosyncrasies of AD embedded in educational videos on the COVID-19 response, released by the New Zealand Government, through a corpus-based approach. Research questions include, in terms of AD in educational videos on COVID-19 response:

- (1) What are the syntactical and grammatical idiosyncrasies?
- (2) What are the idiosyncrasies of lexical choices?
- (3) How do the considerations of cohesion and coherence, text types, and genres shape the textual representations?

Based on the results of syntactical, grammatical, and lexical idiosyncrasies as well as the investigation into cohesion and coherence, text types, and genres, salient features were summarized as the common cores of AD for educational videos in the healthcare context. A clear understanding of the idiosyncrasies and common cores of such a special type of AD enables an effective and efficient presentation for science communication upon the next public health emergency.

Methods

This study has created a corpus of AD products that specifically relate to the topic of COVID-19 educational videos, called the “Audio Description Corpus of COVID-19 Educational Videos” (ADCCEV). A pyramid-shaped matrix named the “Textual and Linguistic Audio Description Matrix” (TLADM) has been drawn on in this study to investigate the unique textual and linguistic characteristics of audio description.

Audio Description Corpus of COVID-19 Educational Videos (ADCCEV)

Piety (2004) pioneered the monomodal corpus-based investigation of AD products by establishing a 23,000-word AD corpus from four films, determining elements in the unique form of language use from structural and functional angles, and revealing that the language for AD purposes is distinct from both spoken and written language. Salway (2007) also investigated the language of AD scripts for 91 films through a corpus-based

automated analysis, yielding certain peculiarities of AD as a special language and the primary types of information offered by AD based on empirical evidence.

Jiménez Hurtado and Seibel (2012) were among the first batch of researchers who initiated multi-semiotic and multimodal corpus-based AD studies. Their multimodal AD corpus included the segmented, tagged meaning-making representations of 300 films in Spanish and 50 in English, French, and German.

Plenty of studies have probed into the characteristics of AD language used in films (Jankowska & Walczak, 2019; Orero & Vilaró, 2012; Puigdomènech, et al., 2010). This leaves a void to investigate the idiosyncrasies of AD embedded in educational videos in a corpus-based manner, especially against the backdrop of the once-in-a-century COVID-19 pandemic.

In this study, a multi-semiotic, multimodal corpus of English-language educational videos on COVID-19 responses and their AD was created. The audio-described educational videos on COVID-19 response were produced by Unite Against COVID-19, sponsored by the New Zealand Government. The videos were downloaded from the official website of the New Zealand Government through the link <https://covid19.govt.nz/languages-and-resources/alternate-formats/audio-description/>. The Corpus—Audio Description Corpus of COVID-19 Educational Videos (ADCCEV)—included 15 videos with a total duration of 14 minutes and 17 seconds. Over three-quarters (79%) of the audiovisual materials were explainers or animated videos, 14% were tutorials or how-to videos, and the remaining 7% was a lecture-based video (as shown in Table 1).

Table 1
Profile of the ADCCEV

Genre	Count	Total Duration	Average Duration
explainer (Animated Videos)	11	9:39:00	3:03:33
Tutorial (How-to Videos)	2	2:26:00	1:13:00
Lecture	1	2:12:00	2:12:00

The videos were transcribed, segmented, and tagged. AD texts were stored in plain text format. Text files also went through segmentation and tagging automatically and manually, based on lexical choices such as parts of speech, syntactical applications, and grammatical usage as well as text types and genres. The audiovisual products experienced marking together with the identification of local and global coherence.

The ADCCEV was processed by Wordless (Version 3.2.0) developed by Ye (2023) for the calculation of syntactical and lexical properties and the generation of Concordance, Wordlist, Collocation, and Keyword.

Textual and Linguistic Audio Description Matrix (TLADM)

Mazur (2022) developed a pyramid-shaped matrix—Textual and Linguistic Audio Description Matrix (TLADM)—to explore the textual and/or linguistic idiosyncrasies of AD. According to this matrix, text types and genres at the top of the pyramid influence lexical, syntactical, and grammatical choices in generating AD. By classifying multimodal text for AD purposes into five types (informative, narrative, expressive, persuasive, and entertaining despite some overlapping boundaries) and analyzing the source text (ST), Mazur (2020) also presented a functional approach to AD in three different layers: contextual, macro-textual, and micro-textual. The model identifies functional priorities in ST to guide the describer's decisions. The contextual and macro-textual analyses help in selecting macro strategies while micro-textual analysis supports micro strategies. In other words,

the AD text in a program should match its style and be suitable for its target audience through different means of words, sentences, and grammatical tools as well as the application of macro and micro strategies under various circumstances.

For example, documentaries on public health emergencies, considering from a macro and contextual perspective, are likely to feature the usage of specialized jargon to name different variants, diseases, and treatments, along with longer, descriptive sentences, because the ST of such healthcare-themed documentaries is largely narrative and informative. In some shots, expressive language dominates in the ST, where shorter, fast-paced lexicon and syntax are applied to help deliver specific feelings or create an atmosphere.

Another example lies in educational or instructional videos for children, which tend to incorporate easy-to-understand lexicon and syntax suited for the intended age range, from a macro and contextual point of view, to convey the message in an intuitive-to-follow manner. To make the videos appealing, entertaining word and sentence choices are also welcomed in some scenarios out of consideration at the micro level.

The corners of the pyramid—text types and genres, lexical choices, and syntax and grammar—interact to ensure cohesive AD text both within and between modes and to make the text coherent for the intended audience.

Results and Discussion

Drawing on the TLADM model, idiosyncrasies of the AD for educational videos on COVID-19 response are explored from linguistic (syntax, grammar, and lexical choices) and textual (cohesion and coherence & text types and genres) perspectives.

Linguistic Idiosyncrasies

Syntax and Grammar

The Mean Sentence or Sentence Segment Length in Tokens refers to the average number of tokens in a sentence or sentence segment. Meanwhile, the Mean Token/Type Length in Syllables/Characters is the average length of a token or token type in terms of syllables or characters.

From a global perspective, as shown in Table 2, the mean sentence length in tokens is 12.089, the mean sentence segment length in tokens is 9.212, the mean token length in syllables is 4.286, and the mean token length in characters is 5.482.

Table 2
Syntactical Properties of the ADCCEV

	Sentence Length in Tokens (Mean)	Sentence Segment Length in Tokens (Mean)	Token Length in Syllables (Mean)	Token Length in Characters (Mean)
1-1	12.111	7.786	4.541	4.964
1-2	11.500	11.500	4.652	5.053
2-1	12.167	9.125	4.082	4.288
2-2	13.000	9.880	4.364	5.028
2-3	10.531	8.425	4.510	5.068
2-4	12.000	11.226	4.261	4.803
2-5	12.571	8.186	4.534	5.028
2-6	11.382	8.600	4.359	5.024
2-7	12.947	9.840	4.207	4.683
2-8	13.613	9.591	3.912	4.616

2-9	12.926	10.265	3.871	4.709
3-1	4.667	4.667	3.964	4.074
3-2	12.600	9.000	4.810	5.141
3-3	12.059	9.318	4.493	5.065
Total	12.089	9.212	4.286	5.482

In terms of grammar, the AD script utilizes a mix of complete sentences and phrases. The majority of the complete sentences are in the Present Simple tense, as seen in No. 1-1's script with the line "A person with white sunscreen under their eyes smiles."

The materials also contain phrases that use ellipsis, for example, "A mosaic of brightly colored cartoon faces, some wearing masks and others not," which appears in the script of No. 1-1. Additionally, the use of present participles ending in "-ing" and the "with" structure (which appears 38 times) are utilized to convey meaning with fewer words.

This is exemplified in No. 2-2's script, which includes "An arrow pointing to the timeline is labeled the usual way," where the present participle "pointing to the timeline" describes the arrow's spatial direction. Similarly, No. 2-3's script includes "An arrow pointing to the circle is labeled immune system," with the present participle modifying the arrow. No. 2-5's script uses "Text pointing towards the cell reads genome sequence of a virus," where the present participle "pointing towards the cell" modifies the text. The use of present participles ending in "-ing" reduces redundancy and avoids extra words constructing attributive clauses. In No. 3-2's script, the "with" structure and a present participle are combined in "The smartphones inside the pocket and handbag of two diners in a restaurant glow yellow with yellow rectangles radiating from them," conveying the necessary meaning with fewer words.

Lexical Choices

The Type-token Ratio measures the ratio of different words in a file to the total number of words. To standardize this ratio, each file is split into sections of 1,000 words, and the ratio is calculated for each section. Then, the standardized ratio is obtained by averaging these ratios with weights assigned based on the number of words in each section.

The corpus ADCCEV consists of 14 AD files with 1,833 types and 3,232 tokens. The total type/token ratio is 0.638, and the standardized one is 0.638, as shown in Table 3.

Table 3

Lexical Properties of the ADCCEV

	Type-token Ratio	Type-token Ratio (Standardized)
1-1	0.761	0.761
1-2	0.826	0.826
2-1	0.712	0.712
2-2	0.575	0.575
2-3	0.525	0.525
2-4	0.497	0.497
2-5	0.506	0.506
2-6	0.530	0.530
2-7	0.667	0.667
2-8	0.512	0.512
2-9	0.521	0.521

3-1	0.964	0.964
3-2	0.730	0.730
3-3	0.600	0.600
Total	0.638	0.638

The animated explainer No. 2-4, which tells what an mRNA vaccine is, has the lowest standardized type-token ratio at 0.497. The lowest figure in the standardized type-token ratio can be interpreted that animated explainers requiring the least complicated words in the AD as verbal means to deliver the message.

Pronouns

The personal pronouns in Modern English consist of a variety of words such as “I,” “you,” “he,” “she,” “it,” “we,” “they,” “them,” “us,” “him,” “her,” “his,” “hers,” “its,” “theirs,” “our,” and “your.” The ADCCEV corpus has provided 133 results for personal pronouns.

A visual representation of the percentage of personal pronouns is depicted in Figure 1. The most common personal pronouns in the ADCCEV corpus, accounting for 18% each, are “it,” “her,” and “his.” In addition, “he” makes up 14% of the total personal pronoun usage, while “she” accounts for 10%. Particularly, there is a complete absence of personal pronouns such as “I,” “we,” “hers,” “theirs,” “our,” and “your.”

The prevalent usage of the third-person narrative in AD materials, particularly in educational videos with animations, results in a significant number of occurrences of pronouns like “it,” “her,” “his,” “he,” and “she.” This can be attributed to the fact that the primary characters in these films are usually human, regardless of their gender. To put it simply, the high frequency of these pronouns is directly related to the third-person narrative used in these materials.

In only one of the videos, namely No. 2-1, which is presented by a virologist and an immunologist, the personal pronoun “you” is utilized (2%). During this lecture, the experts go into detail about how vaccines operate in protecting the human body, the significance of receiving timely immunizations, who benefits most from boosters, the waiting period for boosters after experiencing COVID-19, and the importance of not delaying booster shots. In this instance, filming of actual individuals has been included, which renders the application of the term “you” more appropriate in addressing the audience more directly and prompting them to take action.

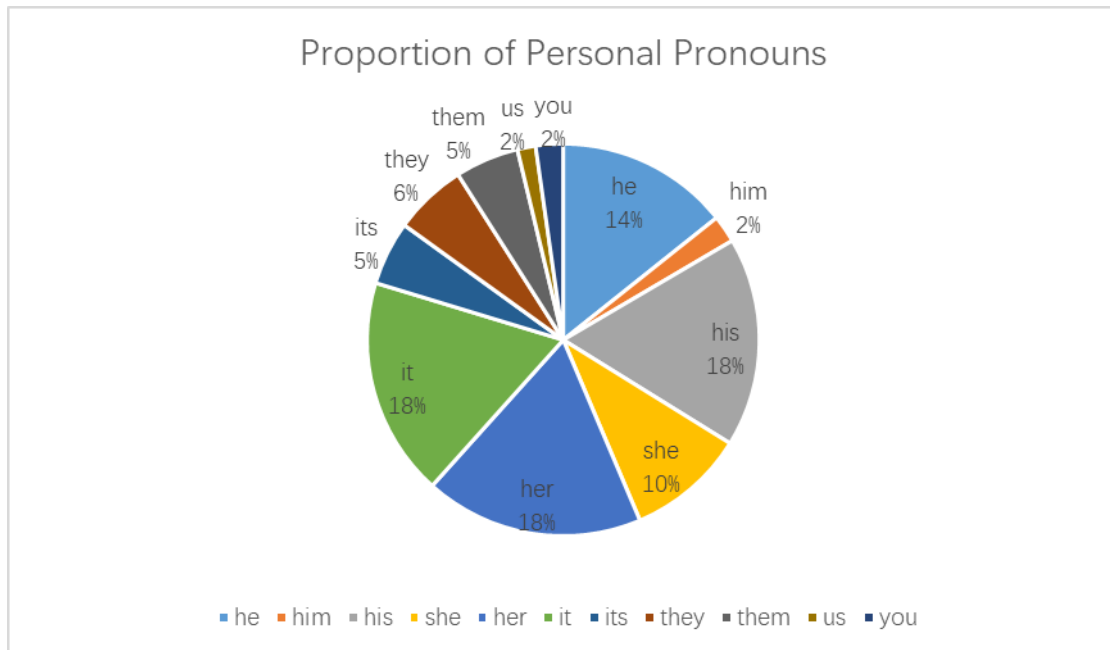


Figure 1. The proportion of personal pronouns in the ADCCEV.

Adjectives

The ADCCEV corpus shows that the colors that occur most often are yellow (24), blue (23), black (18), green (16), orange (12), red (11), white (9), grey (6), brown (3), pink (3), and purple (2).

The color word “yellow” dominates in the usage of colors in ADCCEV, with total occurrences at 24 (Figure 2). What is worth noticing is that “yellow” repeats 6 times with “logo for Unite Against Covid19”, respectively in No. 2-2, 2-3, 2-4, 2-6, 3-2, and 3-3. Subtracting the additional 5 times, “yellow” still leads the frequency of color lexicons used in ADCCEV, at 19 times. The most frequent words following “yellow” in the right context include “logo,” “T-shirt,” “cartoon virus,” “virus,” “rectangles,” “hands,” “background,” “circle,” and “polo shirt” in order of frequency. Yellow is a bright and vibrant color that is often associated with feelings of energy, focus, and stimulation. It is commonly used in various settings and situations where alertness is required. In the corpus, instances including “yellow rectangles,” “a yellow umbrella marked with the NZ VACCINE FACTS,” and “a yellow logo for Unite Against Covid19” demonstrate that the color yellow is used to express urgency and significance.

Number of results: 24

	Left	Node	Right	Sentiment
1	dot govt dot nz slash masks. The	yellow,	black and white logo for Unite Against	-0.056
2	Te Kāwanatanga o Aotearoa. The black and	yellow	logo for Unite Against Covid19. Mā tātau	-0.083
3	and underlines it. The drawing becomes a	yellow	cartoon virus with a fanged face. It	0.000
4	cells wearing hard hats bump into the	yellow	virus, keeping it away from the happy	0.169
5	away from the happy blue cells. The	yellow	virus multiplies, forcing the other cells back.	0.135
6	the circle is labeled immune system. The	yellow	cartoon virus appears. It 's shaped like	0.000
7	Te Kāwanatanga o Aotearoa. The black and	yellow	logo for Unite Against Covid19. Mā tātau	-0.083
8	right and a smaller boy and a	yellow	T - shirt on her left. They eat	0.071
9	Te Kāwanatanga o Aotearoa. The black and	yellow	logo for Unite Against Covid19. Mā tātau	-0.083
10	sandcastle. A woman sits nearby under a	yellow	umbrella labeled with the NZ VACCINE FACTS	0.000
11	VACCINE FACTS logo. A lifeguard in a	yellow	polo shirt and red hat looks out	0.000
12	Te Kāwanatanga o Aotearoa. The black and	yellow	logo for Unite Against Covid19. Mā tātau	-0.083
13	hair wears a green jacket over a	yellow	T - shirt. She passes NOW SHOWING posters	-0.100
14	woman in a green jacket over a	yellow	top waits in a chair. She wears	0.100
15	govt, dot nz slash vaccine facts. The	yellow	and black logo for Unite Against covid19.	-0.083
16	swab is removed. Black text on a	yellow	background. Got symptoms? Get a test. The	-0.083
17	of 2 diners in a restaurant glow	yellow	with yellow rectangles radiating from them. Yellow	0.000
18	diners in a restaurant glow yellow with	yellow	rectangles radiating from them. Yellow hands extend	0.000
19	yellow with yellow rectangles radiating from them.	Yellow	hands extend from the phones and high	0.040
20	were 1.5 m apart for 25 minutes.	Yellow	lines radiate from a larger smartphone, displaying	0.000
21	Te Kāwanatanga o Aotearoa. The black and	yellow	logo for Unite Against Covid19. Covid19 dot	-0.083
22	red people appear. The majority are now	yellow	and have an image of a smartphone	0.000
23	ages. The NZ covid tracer within a	yellow	circle. The logo for the New Zealand	0.068
24	Te Kāwanatanga o Aotearoa. The black and	yellow	logo for Unite Against Covid19. Mā tātau	-0.083

Figure 2. Concordances of “yellow”.

The color word “red” has total occurrences at 11 in the usage of colors in ADCCEV (Figure 3). It appeared the most frequently (at 4 times) in File 3-3, an animated explainer focusing on New Zealand’s COVID Tracer app, specifically how contact tracing keeps people safe. The nouns modified by “red” are mostly geometric figures, such as circles, exclamation points, warning exclamation marks, and x-es. This indicates that in educational videos, the color red can be utilized on circular shapes to signify the importance, urgency, negativity, unhealthiness, and hostility of a given topic.

Number of results: 11

	Left	Node	Right	Sentiment
1	it, causing it to disappear. A large	red	virus appears. None of the antibodies match	0.107
2	to the vaccine 's square shaped spokes.	Red	xes appear beside them. A new antibody	0.068
3	lifeguard in a yellow polo shirt and	red	hat looks out through a pair of	0.000
4	box of popcorn. Inside the movie theater,	red	curtains framed the big screen that shows	0.000
5	a blue sky above a community of	red	brick flats. Find out more covid19 dot	0.125
6	presses the enable Bluetooth tracing button. A	red	exclamation point appears on the phone screen.	0.000
7	other smaller phones, which then all display	red	warning exclamation marks. The logo for the	-0.042
8	surround the woman. One person circle turns	red	and more red circles of people branch	0.167
9	One person circle turns red and more	red	circles of people branch outwards. The circles	0.167
10	order on the app. The circle of	red	people appear. The majority are now yellow	0.000
11	smartphone next to them. One circle is	red.	Wearing a dressing gown and holding a	0.000

Figure 3. Concordances of “red”.

Textual Idiosyncrasies

Cohesion and Coherence

Merely translating images into words is not enough; it is also essential to establish cohesive connections that exist within and between various forms of expression in a film, such as images, sounds, dialogues, and their combinations.

In Video No. 1-1, the sounds of subway brakes and doors opening are accompanied by the following AD reading, “The doors to a commuter train open and the mask conductor inside waves happily to the nervous looking, unmasked woman on the station platform.” The part of AD “unmasked woman on the station platform” precedes a woman without wearing a mask showing in the visual format. The sounds, the AD scripts, and the visual components constitute intermodal cohesion and coherence.

Text Types and Genres

One of the fundamental principles in writing AD script is to ensure that it accurately reflects the mood and style of the movie. A seamless flow that aligns with the intended mood and style of the video is critical for the overall coherence and authenticity of the AD.

In the case of the ADCCEV corpus, the informative and entertaining style of AD is commonly used in the 11 animated explainer videos. Explainer videos are designed to inform or educate the audience about a particular subject and therefore require a straightforward, informative style of AD. This type of AD focuses on engaging information that attracts and retains the audience’s attention.

The two how-to tutorial videos typically require a more narrative style of AD as they take the viewer through a step-by-step process. The AD script needs to create a direction that guides viewers through each step, preventing them from missing out on crucial details.

The lecture-based video requires an expressive and persuasive style of AD. In such videos, the presenter or instructor tries to persuade the audience to take a particular course of action or adopt a specific point of view. The AD script reflects the persuasive nature of the video and creates a sense of urgency that encourages the viewer to take action.

Conclusion

In terms of linguistic idiosyncrasies of the selected educational videos on COVID-19 response, the AD script uses a mix of complete sentences and phrases, the majority being in Present Simple tense. Present participles and the “with” structure are used for brevity. Vocabulary is diverse, with simpler words for animated explainers. Third-person pronouns are common in educational videos. Yellow denotes urgency, and red indicates importance, negativity, and hostility.

From the perspective of textual idiosyncrasies, coherence is achieved through intermodal components that align with the video’s mood and style. AD style varies depending on the video’s purpose, from informative to narrative or expressive.

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